

Carbon waste as graphene

Team of researchers have developed methods for turning carbon waste into graphene.
<https://digit.in/dec21-17>



New plant extract to help lose fat

Mitochondria discovery reveals a plant extract that could combat obesity.
<https://digit.in/dec21-38>

SCIENTISTS DISCOVER THE REASON BEHIND PEARLS' SYMMETRY



Now we know why they are round!

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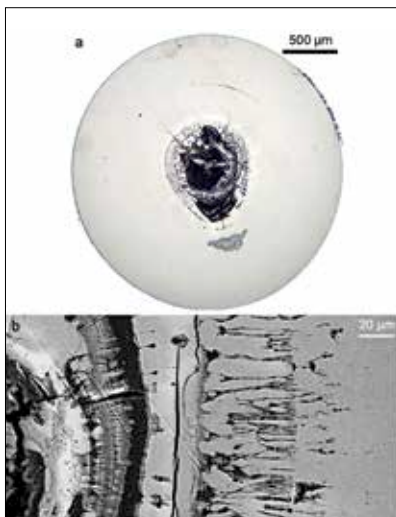
earls are one of the most beautiful naturally occurring materials. Diamonds, heralded to be the

most expensive of all, have to be cut by a master craftsman first, but, when it comes to pearls, they are naturally perfectly shaped and have an outward appearance that is instantly appealing as soon as they come out of the molluscs that birthed them.

While everyone knows where they come from, the mystery behind their shape had remained unsolved. Thanks to the combined efforts of a team of researchers from the University of Michigan, Australian National University, Molecular Foundry, Western Norway University of Applied Sciences, and Cornell University, we now have the answer to this question.

In the research paper titled, "The mesoscale order of nacreous pearls" in

the Proceedings of the National Academy of Sciences, the team published their findings about how they came to a conclusion about the shape of pearls. The formation process is pretty simple to understand. The pearls are formed inside of a mollusc by stacking smooth layers of minerals, protein, and other such materials, collectively called nacre. This nacre, as it keeps on piling up, layer upon layer, eventually falls into symmetry, and this is what results in the perfectly rounded pearl.



Overview of non-bead-cultured "keshi" pearl

RESEARCH PROCESS AND THE FINDINGS

The very process of starting the formation of the pearl is very interesting. As stated in the research paper, "Pearls form either as a natural response to mantle tissue injury or when mantle tissue is deliberately transplanted from a donor into a host animal for pearl culturing." Whatever the case may be, the pearl's first phase of growth occurs as the mantle epithelium grows and develops a closed cyst, also called the pearl sac. This pearl sac is responsible for reproducing the structure of the shell that is there.

The way in which the pearl is round is that instead of arbitrarily stacking layer over layer, forming a one-sided oval-shaped pearl, the mollusc makes sure that it takes corrective action to ensure an equal distribution of nacre on the surface of the pearl. To ensure that the pearl stays in the shape of a sphere, the mollusc adjusts the different aspects of the nacre layer. First up, through a series of internal processes and changes, it makes sure that the